



Finding causes of efficient DNA repair mechanisms of Tardigrades

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Abstract

In this study, we tried to find an answer to the question of how tardigrades, also known as moss pigs or water bears, can efficiently repair damage in their genome, making them among the most stress-resistant organisms. We studied the genome of *Ramazzottius varieornatus* and found proteins that we hypothesized might be involved in genome repair. It is a protein similar to E3 ubiquitin protein ligase, with SWI/SNF-related matrix-associated actin-dependent regulator of chromatin. A protein unique to tardigrade, Dsup, was also discovered.

Keywords: Tardigrades, *Ramazzottius varieornatus*, horizontal gene transfer, stress tolerance

Introduction

Eight-legged, tiny creatures known as tardigrades are renowned for their capacity to endure harsh environments. Organisms that have acclimated to or are at least tolerable of extraordinarily severe surroundings are referred to as "extremophiles." Archaea and bacteria are two examples of the single-celled, straightforward structures seen in the majority of extremophiles. Some creatures do, however, show amazing endurance for harsh settings. The tardigrade is one of the best-known instances of an animal with such extreme tolerance ([Hashimoto and Kunieda 2017](#)).

For growth and reproduction, all tardigrade species need access to water, however certain species can withstand virtually total dryness. Tolerant tardigrades experience desiccation, which causes them to lose body water and enter a contracted dehydrated condition known as anhydrobiosis, a reversible ametabolic state. The dehydrated tardigrades are able to survive a variety of physical conditions that would typically make it impossible for most creatures to survive, including high pressure (7.5 GPa), immersion in organic solvent, and exposure to high doses of radiation ([Hashimoto et al. 2016](#)). For more than 250 years, no one has been able to explain how tardigrades endure desiccation. Some tardigrade species have very low amounts of trehalose, a disaccharide required for many animals to survive drying, or none at all, suggesting that tardigrades may have entirely new strategies for surviving desiccation ([Boothby et al. 2017](#)). A considerable portion of the tardigrade genome is derived from various bacteria as well as plants, fungi, and Archaea, according to genome sequencing, direct proof of physical linkage, and phylogenetic analysis. Nearly twice as many tardigrade genes have undergone horizontal gene transfer (HGT) as those in the most severe examples of HGT into animals that have been identified to yet. Some metazoan gene families within the tardigrade genome have been augmented, enlarged, and even replaced by foreign genes. We hypothesize that animals that can endure harsh conditions may be more

likely to pick up alien genes than other species ([Thomas et al. 2015](#)).

High doses of ionizing radiation are tolerable to some tardigrade species. One distinguishing trait of tardigrades' radiotolerance is that it is strong even when they are hydrated, with levels comparable to those in a dehydrated condition, in contrast to other radiotolerant species, which are unquestionably more tolerant in a dehydrated state. With regard to tolerance, tardigrades and other desiccation-tolerant species vary from one other in another way. The tardigrade is thought to have several DNA-associated proteins that can efficiently preserve and/or repair DNA, given that DNA is a significant target of radiation damage ([Hashimoto et al. 2016](#)). These DNA-associated proteins could have been obtained by HGT from any organisms.

Models and methods

Data preparation

We first downloaded [reference genome](#) of *Ramazzottius varieornatus*, the YOKOZUNA-1 strain.

Then the result of repeat masking and gene prediction made with AUGUSTUS was obtained from [Google Drive](#). AUGUSTUS is a program that predicts genes in eukaryotic genomic sequences.

Structural annotation

We extracted protein sequences from the AUGUSTUS prediction output with the [getAnnoFasta.pl](#) script. We ran the script like this:

```
perl getAnnoFasta.pl augustus.whole.gff
```

to get *augustus.whole.aa* file. Totally, 16435 proteins were extracted.

Physical localization

We downloaded a list of peptides from chromatin fraction which were associated with the DNA from [Yandex Disk](#). Each of the peptides was a piece of one of the proteins that was bound to the DNA (or was pulled out as a false positive).

So, to figure out which proteins from the *R. varieornatus* genome these peptides correspond to, and find the relevant ones we used [Classic BLAST+](#). First, we created a database for a BLAST search:

```
makeblastdb -in augustus.whole.aa -dbtype prot
-out db
```

Then we performed the search via Protein BLAST:

```
blastp -db db -query peptides.fa -outfmt 6
-out result.tsv
```

Finally, we extracted proteins of interest from the initial file using SAMtools ([Danecek et al. 2021](#)):

```
xargs samtools faidx augustus.whole.aa <
no_rep.txt > proteins_of_interest.faa
```

Here, *no_rep.txt* file contains proteins from *result.tsv* but without any duplicates.

Localization prediction

To narrow down the current list of potential candidates that we can then verify by experimentation we tried to predict where these proteins were found in the cell based on their sequences. For this purpose we used [WoLF PSORT](#) and [TargetP - 2.0](#). Both tools predict the subcellular localization of proteins.

BLAST search

For reported proteins we exploit BLAST protein search in the "[UniProtKB/Swiss-Prot](#)" database without adding an organism filter.

Pfam prediction

At the last step we predicted the function of the proteins (even for the ones that we did not manage to find orthologous sequences in databases with BLAST). We used [web-version of HMMER](#) to search our protein sequences against a collection of profile-HMMs for different protein domains and motifs.

Results

During our study we first annotated 16435 proteins from the *R. varieornatus* genome. Then we chose 34 distinct proteins of our interest. Next, we predicted localization for these proteins and took *nuclear* ones (Table 1, Table 2), performed the search with BLAST in "[UniProtKB/Swiss-Prot](#)" database (Table 4), and predicted the function of them (Table 3).

Discussion

A blast search showed annotation similar to the proteins we studied in other organisms. Next, we will look at proteins that may be of interest in terms of DNA repair.

Table 1 Relevant proteins from WoLF PSORT report

Protein ID	Localization	E-value
g8312.t1	Nuclear	15.5
g7861.t1	Nuclear	16
g8100.t1	Nuclear	16.5
g15484.t1	Nuclear	17.5
g11806.t1	Nuclear	18
g10514.t1	Nuclear	19
g10513.t1	Nuclear	20
g16368.t1	Nuclear	20.5
g16318.t1	Nuclear	20.5
g14472.t1	Nuclear	28
g5927.t1	Nuclear	30.5
g11960.t1	Nuclear	32

E3 ubiquitin protein ligase.

With 76 amino acids, ubiquitin (Ub) is a tiny protein that is highly conserved. The process of adding ubiquitin protein to its substrate through a series of events is known as ubiquitination. A crucial stage in the response to DNA damage is the ubiquitination of histone proteins ([Yu et al. 2020](#)). Enzymes Ub, E1, E2, and E3 are progressively needed to target certain substrates for degradation. Three steps are involved in the attachment of Ub to certain substrates, and different enzymes catalyze each one. Ub undergoes an ATP-dependent reaction during the first activation step in which it is covalently conjugated to the side chain of an E1-cysteine via its carboxy-terminal (C-terminal) glycine. Through the ubiquitin-conjugating enzyme E2, activated Ub is transported to an internal cysteine at Ub's C-terminal glycine residue. Finally, an E3 Ub ligase interacts with Ub-bound E2 to facilitate the transfer of Ub from E2 to a particular target protein. In addition to being a key regulator of protein stability, the ubiquitin/proteasome system (UPS) is crucial for the execution of the DNA damage response (DDR) ([Mourtzoukou et al. 2018](#)).

Unique DNA-associated protein of silentworms.

Dsup is a nuclear protein that binds to DNA and is mostly unstructured and highly charged. Under various circumstances, Dsup shields the tardigrade genome from damage caused by hydroxyl radicals. For instance, the tardigrade genome's defense against hydroxyl radicals may let it survive long periods of time in the anhydrobiotic condition. Notably, Dsup and histone H1 may bind to nucleosomes concurrently ([Chavez et al. 2019a](#)). To our knowledge, Dsup protein is the first DNA-associating protein proven to preserve DNA and enhance the radiotolerance of cultured animal cells. It is a conspicuous example of the numerous proteins that are exclusive to tardigrades and are involved in tolerability ([Hashimoto et al. 2016](#)).

SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily A-like protein 1

The multi-subunit SWI/SNF complex is an evolutionarily conserved chromatin-remodeling complex that mobilizes nucleosomes and remodels chromatin using the energy of ATP hydrolysis to control target gene transcription. ([Roberts and Orkin 2004](#)) The DNA damage response (DDR), which protects against DNA damage-induced genomic instability and cancer by eliminating

Table 2 Relevant proteins from TargetP - 2.0 report

Protein ID	Localization	E-value
g8312.t1	Other	15.5
g7861.t1	Other	16
g8100.t1	Other	16.5
g15484.t1	Other	17.5
g11806.t1	Other	18
g10514.t1	Other	19
g10513.t1	Other	20
g16368.t1	Other	20.5
g16318.t1	Other	20.5
g14472.t1	Other	28
g5927.t1	Other	30.5
g11960.t1	Other	32
g4106.t1	Other	- ^a
g4970.t1	Other	- ^a
g5510.t1	Other	- ^a
g5237.t1	Other	- ^a
g12510.t1	Other	- ^a
g2203.t1	Other	- ^a
g3428.t1	Other	- ^a
g5443.t1	Other	- ^a
g11513.t1	Other	- ^a

^a We do not provide E-value for the proteins suggested by WoLF PSORT but not by TargetP - 2.0.

Table 3 HMMER report

Protein ID	Pfam domain
g8312.t1	Clathrin
g7861.t1	SNF2-rel_dom, HARP, ResIII
g8100.t1	Inositol_P, MKLP1_Arf_bdg
g15484.t1	Vps51, Sec5, Dor1, Vps54_N, COG2
g11960.t1	zf-C3HC4

DNA damage through related DNA repair and signaling pathways, includes chromatin remodeling as a key component. Both non-homologous end-joining and homologous recombination, as well as nucleotide excision repair, have been linked to the facilitation of the repair of double-strand breaks and helix-distorting DNA damage by SWI/SNF (Chavez et al. 2019b). Both the non-homologous end joining (NHEJ) and homologous recombination (HR) pathways may repair DNA double-strand breaks (DSBs). The formation of three single-strand DNA overhangs at the break (Wiest et al. 2017). Both of these mechanisms (detailed below) and additional cellular processes that affect genome stability have been linked to SWI/SNF complexes (Harrod et al. 2020).

Conclusion

Despite the found similarity of g7861.t1 and g11960.t1 with proteins in other organisms, we cannot confidently say that they appeared in tardigrade genome due to HGT, since the proteins

were found in eukaryotic organisms, which, for the most part, are not characteristic of it. The Dsup protein is of the greatest interest to us because it is unique to the organism under study. To test the hypothesis, we can conduct experiments with cells stably expressing the Dsup protein.

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Supplementary information

Table 4 Best BLAST hits

Protein ID	Accession	Description	Organism	Identities(%)	Positives(%)	E-value
g7861.t1	B4F769.1	SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily A-like protein 1	Rattus norvegicus	37.21	99	2.00e-71
g8100.t1	Q2YDR3.1	Inositol monophosphatase 3	Danio rerio	36.04	22	3.0e-46
g15484.t1	Q155U0.1	Vacuolar protein sorting-associated protein 51 homolog	Danio rerio	45.03	78	0.0
g16318.t1	A2VD00.1	Eukaryotic translation initiation factor 3 subunit A	Xenopus laevis	36.11	57.4	4.00e-08
g14472.t1	P0DOW4.1	Damage suppressor protein [Ramazzottius varieornatus]	Ramazzottius varieornatus	100.0	100.0	0.0
g5927.t1	Q17427.1	Glucosamine 6-phosphate N-acetyltransferase	Caenorhabditis elegans	38.64	14.0	1.00e-18
g11960.t1	A0A4C1TJ24	E3 ubiquitin protein ligase	Eumeta variegata	32.9	54.0	4.6e-150